```
clc;
clear;
close all;

% ============================================================
% IIR FILTER COEFFICIENTS
% ============================================================
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

N = 50;
x = [1 zeros(1,N-1)];   % Unit impulse input
n = 0:N-1;

% ============================================================
% 1. DIRECT FORM-I AND DIRECT FORM-II
% ============================================================
% Transfer-function reference response
y1 = filter(b,a,x);

% Direct Form-II Transposed realization
y2 = filter(b,a,x);

% ============================================================
% 2. CASCADE REALIZATION
% ============================================================
[sos,g] = tf2sos(b,a);

yc = x;

for k = 1:size(sos,1)
    yc = filter(sos(k,1:3),sos(k,4:6),yc);
end

yc = g*yc;

% ============================================================
% 3. PARALLEL REALIZATION
% ============================================================
[r,p,k] = residuez(b,a);

% Direct polynomial term
yp = filter(k,1,x);

% Add partial-fraction terms
for i = 1:length(r)
    yp = yp + filter(r(i),[1 -p(i)],x);
end

% ============================================================
% 4. DISPLAY FILTER COEFFICIENTS
% ============================================================
```

```
disp('Numerator coefficients b =');
disp(b);

disp('Denominator coefficients a =');
disp(a);

disp('Second Order Sections (Cascade) =');
disp(sos);

disp('Cascade Gain =');
disp(g);

disp('Residues (Parallel) =');
disp(r);

disp('Poles (Parallel) =');
disp(p);

disp('Direct term =');
disp(k);

% ============================================================
% 5. ERROR AND PERCENTAGE DEVIATION
% ============================================================
E1 = max(abs(y1-y2));
E2 = max(abs(y1-yc));
E3 = max(abs(y1-yp));

reference = max(abs(y1));

if reference > 0
    D1 = E1/reference*100;
    D2 = E2/reference*100;
    D3 = E3/reference*100;
else
    D1 = 0;
    D2 = 0;
    D3 = 0;
end

fprintf('\n=======================================\n');
fprintf('IIR REALIZATION COMPARISON\n');
fprintf('=======================================\n');

fprintf('DF-I vs DF-II Maximum Error = %.8e\n',E1);
fprintf('DF-I vs Cascade Maximum Error = %.8e\n',E2);
fprintf('DF-I vs Parallel Maximum Error = %.8e\n',E3);

fprintf('\nPercentage Deviation:\n');
fprintf('DF-I vs DF-II = %.8e %%\n',D1);
fprintf('DF-I vs Cascade = %.8e %%\n',D2);
fprintf('DF-I vs Parallel = %.8e %%\n',D3);

% ============================================================
```

```
% 6. IMPULSE RESPONSE COMPARISON
% ========================================================
figure('Name','Impulse Response Comparison');

stem(n,y1,'filled');
hold on;
stem(n,y2,'--');
stem(n,yc,':','LineWidth',1.5);
stem(n,yp,'-.');

grid on;
xlabel('Sample Index (n)');
ylabel('Amplitude');
title('Impulse Response of IIR Realizations');

legend('Direct Form-I','Direct Form-II', ...
       'Cascade','Parallel','Location','best');
hold off;

% ========================================================
% 7. NUMERICAL ERROR PLOT
% ========================================================
figure('Name','Numerical Error');

plot(n,abs(y1-y2),'--','LineWidth',1.2);
hold on;
plot(n,abs(y1-yc),':','LineWidth',1.5);
plot(n,abs(y1-yp),'-.','LineWidth',1.2);

grid on;
xlabel('Sample Index (n)');
ylabel('Absolute Error');
title('Numerical Error Between Realizations');

legend('DF-I vs DF-II','DF-I vs Cascade', ...
       'DF-I vs Parallel','Location','best');
hold off;

% ========================================================
% 8. FREQUENCY RESPONSE
% ========================================================
figure('Name','Frequency Response');

freqz(b,a,1024);
title('Frequency Response of IIR Filter');

% ========================================================
% 9. POLE-ZERO PLOT
% ========================================================
figure('Name','Pole-Zero Plot');

zplane(b,a);
title('Pole-Zero Plot of IIR Filter');
```

```
Numerator coefficients b =
    0.0048    0.0193    0.0289    0.0193    0.0048

Denominator coefficients a =
    1.0000   -2.3695    2.3140   -1.0547    0.1874

Second Order Sections (Cascade) =
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327

Cascade Gain =
    0.0048

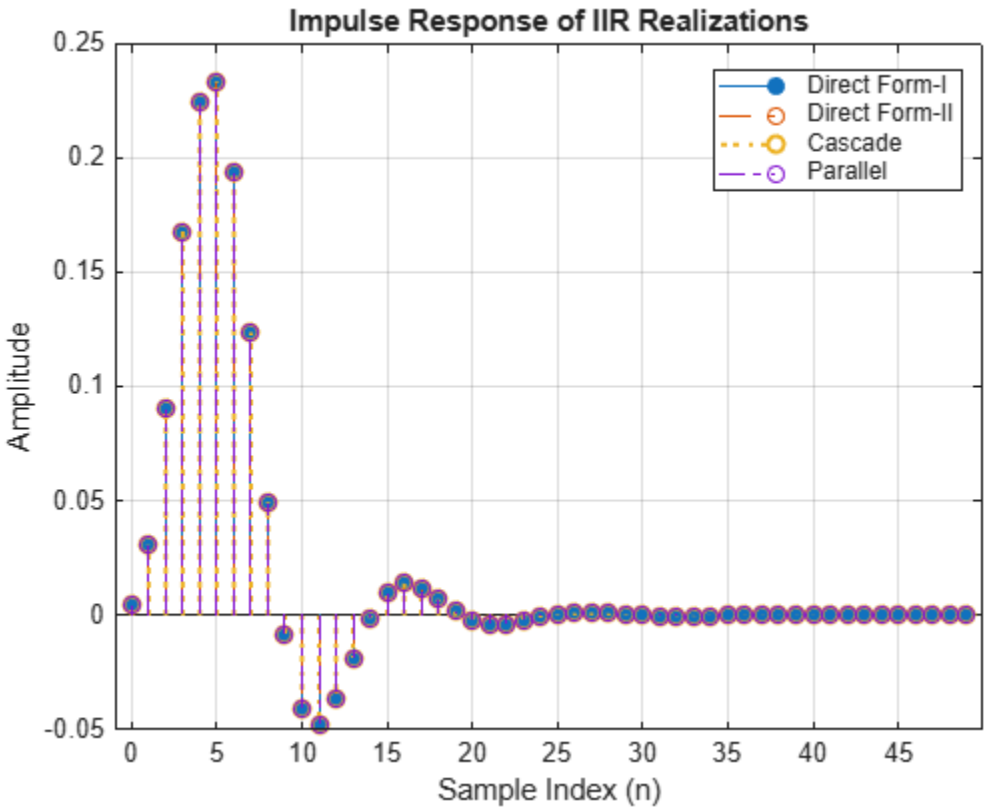
Residues (Parallel) =
  -0.2698 + 0.1343i
  -0.2698 - 0.1343i
   0.2594 - 0.8024i
   0.2594 + 0.8024i

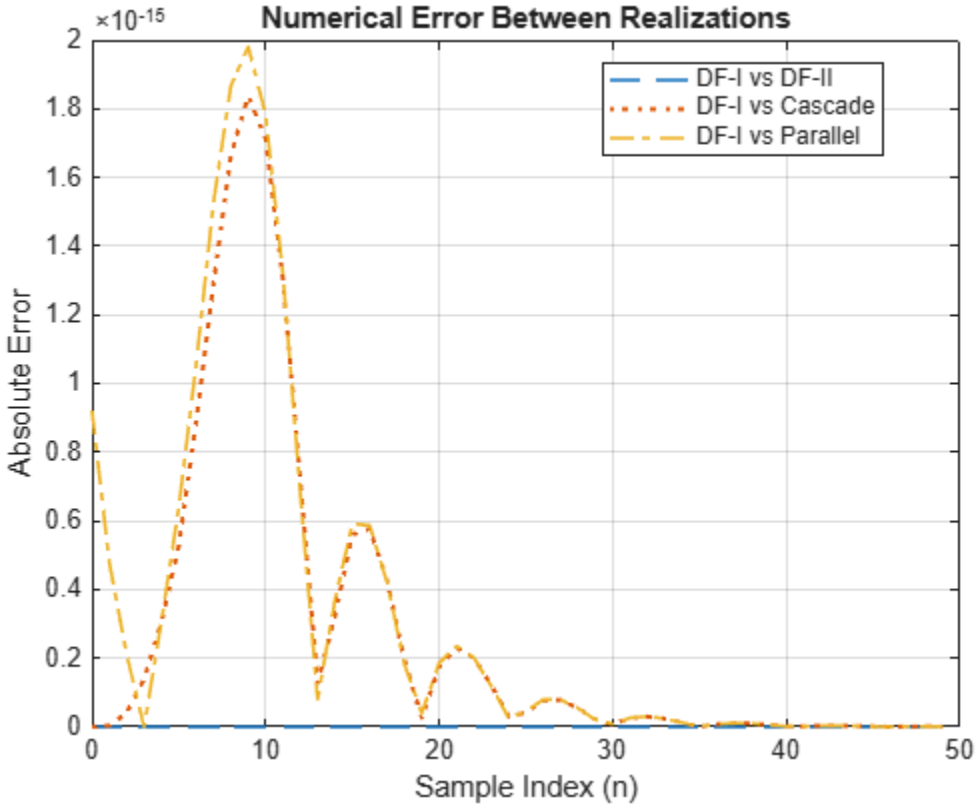
Poles (Parallel) =
   0.6604 + 0.4433i
   0.6604 - 0.4433i
   0.5243 + 0.1459i
   0.5243 - 0.1459i

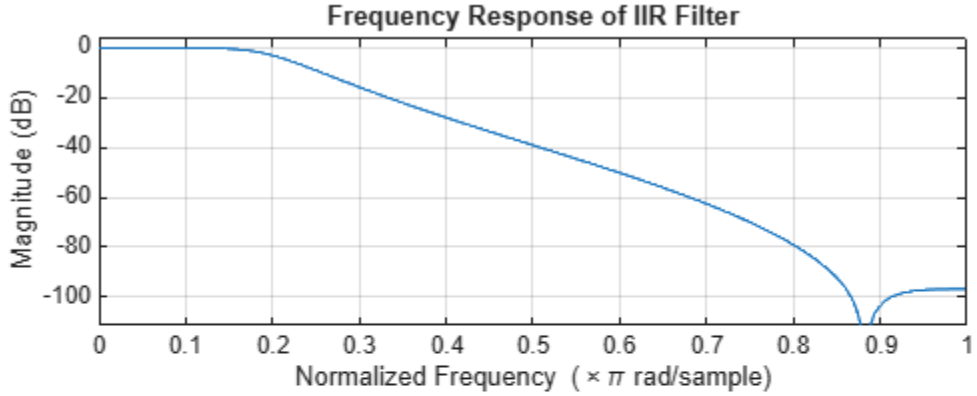
Direct term =
    0.0256

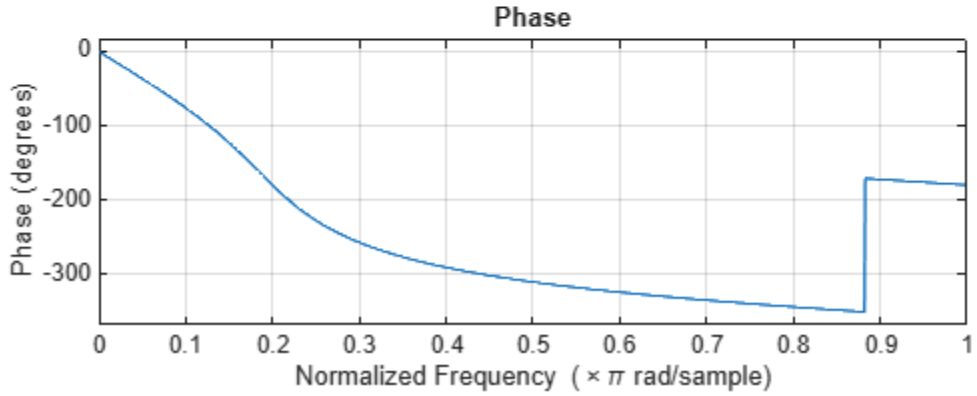

=======================================
IIR REALIZATION COMPARISON
=======================================
DF-I vs DF-II Maximum Error = 0.00000000e+00
DF-I vs Cascade Maximum Error = 1.83707216e-15
DF-I vs Parallel Maximum Error = 1.98105421e-15

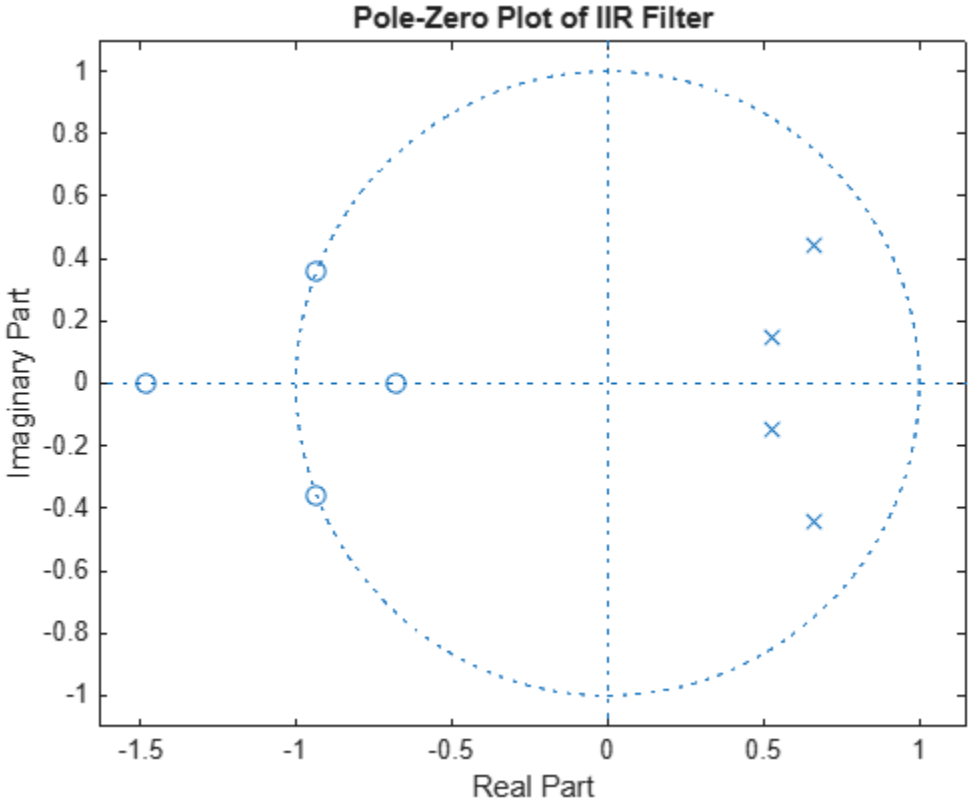
Percentage Deviation:
DF-I vs DF-II = 0.00000000e+00 %
DF-I vs Cascade = 7.87747246e-13 %
DF-I vs Parallel = 8.49487587e-13 %
Warning: Using only the real component of complex data.
```


Impulse Response of IIR Realizations
Direct Form-I
Direct Form-II
Cascade
Parallel
Amplitude
Sample Index (n)
0.25
0.2
0.15
0.1
0.05
0
-0.05
0
5
10
15
20
25
30
35
40
45


Numerical Error Between Realizations
×10-15
DF-I vs DF-II
DF-I vs Cascade
DF-I vs Parallel
Absolute Error
Sample Index (n)
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
10
20
30
40
50


Frequency Response of IIR Filter
Magnitude (dB)
0
-20
-40
-60
-80
-100
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)


Phase
Phase (degrees)
0
-100
-200
-300
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)


Pole-Zero Plot of IIR Filter
Imaginary Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.5
-1
-0.5
0
0.5
1
Real Part

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